Interface summary

21 interfaces and 6 type aliases from 8 files, in 6 groups of related interfaces. Inherited members are listed under the interface that declares them. Generated 2026-11-10 by npm run summary.

RectangleOps group

Point geometry.ts

A location, in the coordinate space its use states.

Used by RectangleOps; Shape, InkStroke, Circle (page 2); ShapeLocatorQueries (page 3); RenderGroup (page 4); EventDistributor (page 5).

```
readonly x: number
readonly y: number
```

Rectangle geometry.ts

An axis-aligned rectangle covering the half-open region [x, x + width) × [y, y + height).

Used by RectangleOps; Shape (page 2); ShapeLocatorQueries (page 3); RenderGroup (page 4); LayoutObject (page 5).

```
readonly x: number
readonly y: number
readonly width: number
readonly height: number
```

RectangleOps geometry.ts

Operations on rectangles, following Rectangle's half-open semantics. cut, splitHoriz, splitVert, inset and place are each defined exactly by their reference functions in geometry.vocab.ts.

```
contains(r: Rectangle, p: Point): boolean
```
Whether p lies in r.

```
intersects(a: Rectangle, b: Rectangle): boolean
```
Whether a and b overlap.

```
cut(r: Rectangle, edge: Edge, size: number): [Rectangle,
  Rectangle]
```
Cuts a strip of the given size off one edge of r.

```
splitHoriz(r: Rectangle, ...weights: number[]):
  Rectangle[]
```
Splits r left to right into one rectangle per weight, each as wide as its share of r's width, and as tall as r. splitHoriz(r, 5, 10, 5) gives a quarter, a half and a quarter.

```
splitVert(r: Rectangle, ...weights: number[]):
  Rectangle[]
```
Splits r top to bottom into one rectangle per weight, each as tall as its share of r's height, and as wide as r.

RectangleOps, continued

```
inset(r: Rectangle, d: number): Rectangle
```
r shrunk by d on every side.

```
place(r: Rectangle, size: { readonly width: number;
  readonly height: number }, alignX: Align, alignY:
  Align): Rectangle
```
A rectangle of the given size placed within r, aligned along each axis at the start, the center (rounded down to a whole unit) or the end.

Viewport geometry.ts

A 2D similarity transform (uniform scale, rotation, translation) from canonical coordinates to view coordinates (see the top of this file).

Used by View (page 4).

```
readonly scale: number
```
View units per canonical unit.

```
readonly rotation: number
```
Radians.

```
readonly tx: number
readonly ty: number
```

Types

```
type Edge = "top" | "bottom" | "left" | "right"
```
A side of a rectangle.

Used by RectangleOps.

```
type Align = "start" | "center" | "end"
```
Where a smaller rectangle goes within a larger one, along one axis.

Used by RectangleOps.

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Shape group

Refers to Point, Rectangle (page 1).

Color color.ts

An sRGB color with straight (not premultiplied) alpha.

Used by InkStroke, Circle.

```
readonly r: number
readonly g: number
readonly b: number
readonly a: number
```

Shape shape.ts

A region of the plane, such as an ink stroke.

Used by InkStroke, Circle, ShapeCollection; ShapeLocator (page 3); ShapeView, DrawingGroup (page 4).

```
readonly kind: string
```
Identifies the shape's type.
```
boundingBox(): Rectangle
```
The exact extents of the shape: x and y are its minimum x and y, and x + width and y + height are its maximum x and y.
```
intersects(r: Rectangle): boolean
```
Whether at least one point lies both in this shape and in r.
```
contains(p: Point): boolean
```
Whether p lies in the shape.

InkStroke extends Shape shape.ts

A stroke of digital ink: a polyline drawn with a round pen.

Extends Shape.

```
readonly kind: "ink"
readonly points: readonly Point[]
```
The polyline's vertices, in drawing order.
```
readonly thickness: number
```
Pen diameter, in canonical units.
```
readonly color: Color
```

Circle extends Shape shape.ts

A filled circle: the points within radius of center, with the half-open rule at its maximum x and y.

Extends Shape.

```
readonly kind: "circle"
```

Circle, continued

```
readonly center: Point
readonly radius: number
```
In canonical units.
```
readonly color: Color
```

ShapeCollection<S extends Shape> shape.ts

A collection of shapes, kept in z-order.

```
add(shape: S): ShapeId
```
Adds shape and returns its new ID.
```
remove(id: ShapeId): boolean
```
Removes the shape with this ID.
```
removeMultiple(ids: readonly ShapeId[]): number
```
Removes every listed ID that is present. ids may contain duplicates and IDs that aren't present; both are ignored.
```
get(id: ShapeId): S | undefined
```
The shape with this ID, or undefined if no such shape exists.
```
readonly size: number
```
The number of shapes in the collection.
```
forEach(visit: (id: ShapeId, shape: S) => void): void
```
Calls visit for each shape, bottom to top in z-order. visit must not modify the collection.
```
compareZ(a: ShapeId, b: ShapeId): number
```
Compares the z-order of two shapes.

Types

```
type ShapeId = number & { readonly __brand: "ShapeId" }
```
Opaque identifier for a shape in a ShapeCollection.

Used by ShapeCollection, ShapeId; ShapeLocatorQueries, ShapeLocator (page 3).

Interface summary

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ShapeLocatorQueries group

Refers to Point, Rectangle (page 1); ShapeId, Shape (page 2).

ShapeLocatorQueries shape.ts

Finds shapes by location.

Used by ShapeLocator.

`inRect(r: Rectangle, out: ShapeId[]): number`
Finds the shapes that intersect r.

`atPoint(p: Point, out: ShapeId[]): number`
Finds the shapes that contain p.

ShapeLocator extends `ShapeLocatorQueries` shape.ts

A spatial index that finds shapes by location, such as a quadtree.

Extends ShapeLocatorQueries.

`add(id: ShapeId, shape: Shape): void`
Indexes shape under id.

`remove(id: ShapeId): boolean`
Removes id from the index.

`removeMultiple(ids: readonly ShapeId[]): number`
Removes every listed ID that is present. ids may contain duplicates and IDs that aren't present; both are ignored.

FrameHandler group

FrameHandler frames.ts

Something that does its work during a display frame.

Used by FrameScheduler; View (page 4).

`frame(timeMs: number): void`
timeMs is the frame's timestamp in milliseconds, on the window's monotonic clock.

FrameScheduler frames.ts

Calls handlers before the window's next display refresh.

`requestFrame(handler: FrameHandler): void`
Asks for one call to handler.frame() before the next display refresh.

Interface summary

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RenderGroup group

Refers to Point, Rectangle, Viewport (page 1); Shape (page 2); FrameHandler (page 3).

EventEmitter<Events extends EventMap> events.ts

Something that emits events.

Used by RenderGroup, View.

```
on<E extends keyof Events & string>(event: E, listener:
   (...args: Events[E]) => void): this
```
Registers listener for event.

```
off<E extends keyof Events & string>(event: E, listener:
   (...args: Events[E]) => void): this
```
Removes the most recent registration of listener for event; does nothing if it has none.

RenderGroup<T> extends EventEmitter<{ change: [] }> view.ts

One group of a scene: items in z-order, found by location, emitting change when they change.

Extends EventEmitter. **Used by** Scene, DrawingGroup.

```
get(id: ItemId): T | undefined
```
The item with this ID, or undefined if no such item is present.

```
readonly size: number
```
The number of items in the group.

```
forEach(visit: (id: ItemId, item: T) => void): void
```
Calls visit for each item, bottom to top in z-order. visit must not change the source.

```
compareZ(a: ItemId, b: ItemId): number
```
Compares the z-order of two items.

```
inRect(r: Rectangle, out: ItemId[]): number
```
Finds the items whose regions intersect r.

```
atPoint(p: Point, out: ItemId[]): number
```
Finds the items whose regions contain p.

DrawingGroup extends RenderGroup<Shape> shapeView.ts

A group of shapes with edits.

Extends RenderGroup.

```
add(shape: Shape): ItemId
```
Adds shape as the top item, and emits change.

```
remove(id: ItemId): boolean
```
Removes the item with ID id, if present, and emits change if it was.

DrawingGroup, continued

```
removeMultiple(ids: readonly ItemId[]): number
```
Removes every listed item that is present, and emits change once if any was.

Scene<T> view.ts

What views draw: groups in canonical coordinates, the same for every view, with no viewport. groups[0] is the bottom group.

Used by View.

```
readonly groups: readonly RenderGroup<T>[]
```

View<T> extends FrameHandler, EventEmitter<{ viewport: [Viewport] }> view.ts

Shows a scene on screen through a viewport, and stays current.

Extends EventEmitter; FrameHandler (page 3). **Used by** ShapeView.

```
readonly scene: Scene<T>
```

```
viewport: Viewport
```
Setting this redraws the view at the next frame and emits viewport with the new value.

```
dispose(): void
```
Ends the view: removes its listeners from the scene's groups, and requests, draws and emits nothing more.

ShapeView extends View<Shape> shapeView.ts

A view of shapes.

Extends View.

```
frame(timeMs: number): void
```
Draws the scene as it is now.

Types

```
type EventMap = Record<string, unknown[]>
```
Each event's name, mapped to the arguments its listeners get.
Used by EventEmitter.

```
type ItemId = number & { readonly __brand: "ItemId" }
```
Identifies an item within one group.
Used by RenderGroup, DrawingGroup, ItemId.

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LayoutObject group

Refers to Rectangle, Point (page 1).

LayoutObject layout.ts

Something placed in a region of a window.

Used by EventDistributor.

```
region: Rectangle
```

Where it is, in the window's coordinates.

EventDistributor<M> extends LayoutObject layout.ts

Passes a platform's positional events to listeners by where they fall.

Extends LayoutObject.

```
locate(event: M[keyof M]): Point
```

Where event happened, in the window's coordinates.

```
on<E extends keyof M & string>(eventName: E, listener:
   Listener<M, E>, target: LayoutObject): void
```

Registers listener for eventName on target.

```
off<E extends keyof M & string>(eventName: E, listener:
   Listener<M, E>, target: LayoutObject): void
```

Removes the most recent registration of listener for eventName on target; does nothing if there is none.

```
distribute<E extends keyof M & string>(eventName: E,
   event: M[E], except?: LayoutObject): boolean
```

Passes event to the innermost target, among those registered for eventName other than except, that contains it.

Types

```
type Listener<M, E extends keyof M> = (eventName: E,
   event: M[E]) => void
```

Gets one of a platform's events: its name, and the platform's event object.

Used by EventDistributor.